

Extra study and consolidation in Algebra

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1 What is this?

Due to my being very interested in algebra, and interested in becoming very good at it, I have these extra notes for me to work through a lot of problems. Basically, there will probably not be any pedagogical value here.

2 Repeating some stuff about ideals, exact sequences and so forth

Problem 2.1. Let

$$R = \frac{k[x, y]}{(xy)}$$

and write $\bar{x}, \bar{y}$ for the residue classes.

What is $(\bar{x})$ as an ideal of R ?

Compute

$$R/(\bar{x}), \quad R/(\bar{y}), \quad R/(\bar{x}, \bar{y}).$$

Then classify them as prime/maximal/neither.

Sol. We note that the ideal generated by $\bar{x} = x + (xy)$ is

$$(\bar{x}) = \{\overline{xf} \mid f \in k[x, y]\}.$$

We have $(xy) \subseteq (x)$ and $(xy) \subseteq (y)$ as ideals of $k[x, y]$.

Consider the projection $\pi : k[x, y] \rightarrow R$ by

$$\pi(a) = \bar{a} = a + (xy).$$

Since π is surjective and $\{x\} \subseteq k[x, y]$ we have

$$\pi[(x)] = (\pi x) = (\bar{x}),$$

(note: surjectivity guarantees that scalar multiplication in R can be pulled back to $k[x, y]$, ensuring $\pi[(x)] \supseteq (\pi(x))$) but also

$$\pi[(x)] = \frac{(x)}{(xy)}$$

since $(xy) \subseteq (x)$.

Similarly,

$$\frac{(y)}{(xy)} = \pi[(y)] = (\bar{y}),$$

and

$$\frac{(x, y)}{(xy)} = \pi[(x, y)] = (\bar{x}, \bar{y}).$$

Thus we can apply the third isomorphism theorem to get

$$\frac{R}{(\bar{x})} = \frac{k[x, y]/(xy)}{(x)/(xy)} \cong \frac{k[x, y]}{(x)} \cong k[y],$$

and

$$\frac{R}{(\bar{y})} = \frac{k[x, y]/(xy)}{(y)/(xy)} \cong \frac{k[x, y]}{(y)} \cong k[x],$$

and

$$\frac{R}{(\bar{x}, \bar{y})} = \frac{k[x, y]/(xy)}{(x, y)/(xy)} \cong \frac{k[x, y]}{(x, y)} \cong k.$$

The first two are integral domains so $(\bar{x})$, $(\bar{y})$ are prime, though not maximal, while $(\bar{x}, \bar{y})$ is maximal since $R/(\bar{x}, \bar{y}) \cong k$ is a field. ■

Problem 2.2. Let

$$R = \frac{k[x, y]}{(xy)},$$

and let

$$\pi : k[x, y] \rightarrow R$$

be the quotient map.

Prove that every prime ideal $P \subseteq R$ contains $\bar{x}$ or $\bar{y}$.

Then, classify every prime ideal of R .

You may use without proof:

$$P \subseteq R \text{ prime} \Leftrightarrow \pi^{-1}(P) \subseteq k[x, y] \text{ is prime containing } (xy).$$

Proof. We have $\bar{x}\bar{y} = 0$, and since P is an ideal, and thus a group, we have $0 \in P$, and since P is prime we must then have

$$\bar{x} \in P \text{ or } \bar{y} \in P.$$

Suppose P contains $\bar{x}$. Then, $(\bar{x}) \subseteq P$.

Then, by the third isomorphism theorem,

$$\frac{R/(\bar{x})}{P/(\bar{x})} \cong R/P,$$

but

$$\frac{R}{(\bar{x})} \cong k[y]$$

so

$$R/P \cong \frac{k[y]}{P/(\bar{x})}.$$

Since P is a prime ideal of R , the quotient R/P is an integral domain. Therefore, $P/(\bar{x})$ must be a prime ideal of $k[y]$.

Since $k[y]$ is a Principal Ideal Domain (PID), its prime ideals are:

1. The zero ideal (0) , or
2. Irreducible polynomial ideals $(f(y))$, where $f(y) \in k[y]$ is an irreducible polynomial.

Under the canonical quotient map $R \rightarrow R/(\bar{x}) \cong k[y]$, these correspond precisely to the prime ideals of R containing $\bar{x}$:

1. $P = (\bar{x})$, corresponding to $(0) \subset k[y]$.
2. $P = (\bar{x}, f(\bar{y}))$, where $f(y) \in k[y]$ is irreducible.

By symmetric reasoning, if $\bar{y} \in P$, then $P/(\bar{y})$ is a prime ideal of $k[x]$, which yields:

1. $P = (\bar{y})$, corresponding to $(0) \subset k[x]$.
2. $P = (\bar{y}, g(\bar{x}))$, where $g(x) \in k[x]$ is irreducible.

Combining both cases, the prime ideals of R consist of:

- Minimal prime ideals: $(\bar{x})$ and $(\bar{y})$.
- Maximal ideals: $(\bar{x}, f(\bar{y}))$ for irreducible $f(y) \in k[y]$, and $(\bar{y}, g(\bar{x}))$ for irreducible $g(x) \in k[x]$.



Problem 2.3. Let R be a commutative ring, I an ideal of R , and M an R -module. You have three objects

$$IM, M, M/IM.$$

Part A.

Identify the two most natural R -linear maps between these objects and use them to construct a short exact sequence.

Give the maps explicitly and check exactness at every nonzero term.

Part B.

Now consider R/I as an R -module.

There is a canonical isomorphism

$$M \otimes_R R/I \cong M/IM.$$

Answer:

- What should the natural map

$$M \times R/I \rightarrow M/IM$$

probably be?

- Why is there an immediate well-definedness issue?
- What property would allow that bilinear map to induce a map from

$$M \otimes_R R/I?$$

Part C.

Suppose $N \subseteq M$. There is an obvious short exact sequence

$$0 \rightarrow N \rightarrow M \rightarrow M/N \rightarrow 0.$$

Tensor all with R/I .

You might hope to get

$$0 \rightarrow N \otimes_R R/I \rightarrow M \otimes_R R/I \rightarrow (M/N) \otimes_R R/I \rightarrow 0.$$

Is that always exact?

Proof of A. IM is the collection of finite sums of $i_j m_j$ where $i_j \in I$ and $m_j \in M$. M/IM is then the collection of

$$\bar{x} = x + IM, \quad x \in M,$$

meaning the residue classes of elements in M modulo IM .

The most natural map from M to M/IM is the projection $\pi : M \rightarrow M/IM$,

$$\pi(x) = \bar{x}$$

sending elements x into their residue classes.

The natural map from IM to M is the inclusion $\iota : IM \rightarrow M$ simply sending

$$\iota(x) = x$$

since $IM \subseteq M$.

R -linearity of ι is essentially trivial since

$$\iota(rx + y) = rx + y = r\iota(x) + \iota(y).$$

For π observe that

$$\begin{aligned}
 \pi(rx + y) &= \overline{rx + y} \\
 &= (rx + y) + IM \\
 &= (rx + IM) + y + IM \\
 &= r(x + IM) + (y + IM) \\
 &= r\bar{x} + \bar{y} \\
 &= r\pi(x) + \pi(y).
 \end{aligned}$$

This gives a natural sequence

$$0 \rightarrow IM \xrightarrow{\iota} M \xrightarrow{\pi} M/IM \rightarrow 0.$$

Then

$$\ker \pi = IM$$

since elements in IM are 0 modulo IM when you take the quotient, and $IM \subseteq M$.

Then

$$\operatorname{im} \iota = IM,$$

since ι maps x to x with $IM \subseteq M$.

We have $\operatorname{im} \pi = M/IM$ since a residue class $\bar{x} \in M/IM$ has at least one representative $x \in M$.

The only map from $M/IM \rightarrow 0$ is the trivial map, so

$$\ker(M/IM \rightarrow 0) = M/IM.$$

For the other side of the sequence there is once again only the trivial map from $0 \rightarrow IM$ with $\operatorname{im}(0 \rightarrow IM) = 0$, and

$$\ker \iota = 0,$$

since $\iota(x) = 0_M$ implies $x = 0_M$, but as $IM \subseteq M$ we have that $0_{IM} = 0_M$.

Thus we have shown exactness at each point in the sequence, so this short sequence is exact. ■

Proof of B. A natural map $\beta : M \times R/I \rightarrow M/IM$ might be

$$(m, \bar{x}) \mapsto \overline{mx},$$

meaning

$$(m, r + I) \mapsto (mr) + IM$$

Notice that $(m, \bar{x}) \mapsto \overline{m + x}$ would very obviously create issues if we try to apply it to the tensor product since we require that $m \otimes 0$ is 0. Furthermore, $m + r$ is generally not an

element of M for arbitrary $r \in R$.

For well-definedness suppose $r + I = s + I$. Then $r - s \in I$.

Then $(r - s)m \in IM$ since $r - s \in I$ and $m \in M$.

Thus we have that

$$\begin{aligned}\beta(m, r) - \beta(m, s) &= (mr + IM) - (ms + IM) \\ &= (m(r - s)) + IM,\end{aligned}$$

but $m(r - s) \in IM$ so this is 0_{IM} .

Lastly

$$\begin{aligned}\beta(m_1 + m_2, \bar{r}) &= ((m_1 + m_2)r) + IM \\ &= (m_1r + m_2r) + IM \\ &= (m_1r + IM) + (m_2r + IM) \\ &= \beta(m_1, \bar{r}) + \beta(m_2, \bar{r}),\end{aligned}$$

and

$$\begin{aligned}\beta(m, \bar{r} + \bar{s}) &= (m(r + s)) + IM \\ &= (mr + ms) + IM \\ &= (mr + IM) + (ms + IM) \\ &= \beta(m, \bar{r}) + \beta(m, \bar{s}),\end{aligned}$$

and

$$\begin{aligned}\beta(am, \bar{r}) &= ((am)r) + IM \\ &= (m(ar)) + IM \\ &= \beta(m, a\bar{r}).\end{aligned}$$

Thus, by the universal property of the tensor product, we can factor through a unique map so that

$$\begin{array}{ccc} M \times R/I & \xrightarrow{\beta'} & M/IM \\ & \searrow & \uparrow \exists! \\ & & M \otimes_R R/I \end{array}$$

commutes. ■

Proof of C. We begin with the short exact sequence

$$0 \rightarrow N \xrightarrow{\iota} M \xrightarrow{\pi} M/N \rightarrow 0.$$

Tensoring with R/I induces maps $\iota_R : N \otimes_R R/I \rightarrow M \otimes_R R/I$ sending

$$n \otimes \bar{r} \mapsto n \otimes \bar{r}$$

by

$$\iota_R(n \otimes \bar{r}) = \iota(n) \otimes r,$$

and $\pi_R : M \otimes_R R/I \rightarrow (M/N) \otimes_R R/I$ sending

$$m \otimes \bar{r} \mapsto \overline{m} \otimes \bar{r}$$

by

$$\pi_R(m \otimes \bar{r}) = \pi(m) \otimes \bar{r} = (m + N) \otimes \bar{r}.$$

In general tensoring is right exact, meaning

$$M' \otimes L \rightarrow M'' \otimes L \rightarrow M''' \otimes L \rightarrow 0$$

would remain exact if

$$M' \rightarrow M'' \rightarrow M''' \rightarrow 0$$

was.

Take for instance

$$0 \rightarrow 2\mathbb{Z} \rightarrow \mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z} \rightarrow 0$$

and tensor with $\mathbb{Z}/2\mathbb{Z}$.

Then we imagine we get an exact sequence

$$0 \rightarrow 2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \rightarrow \mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \rightarrow 0.$$

However, there is no such exact sequence, with the

$$0 \rightarrow 2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z}$$

term being the problem.

Consider $2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2 \rightarrow \mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2$. Take $2 \otimes \bar{1}$. Then the image is $2 \otimes \bar{1}$, but in $\mathbb{Z} \otimes \mathbb{Z}/2$ we have

$$(2 \cdot 1) \otimes \bar{1} = 1 \otimes (2\bar{1}) = 0.$$

In $2\mathbb{Z} \otimes \mathbb{Z}/2$, however, $2 \otimes \bar{1}$ is nonzero. Take the $\mathbb{Z}$ -bilinear map $B : 2\mathbb{Z} \times \mathbb{Z}/2 \rightarrow \mathbb{Z}_2$ given by $B(2m, \bar{n}) = mn \pmod{2}$ and take the unique map factoring through the tensor product $\beta : 2\mathbb{Z} \otimes \mathbb{Z}/2 \rightarrow \mathbb{Z}_2$.

As an aside, the map B is well defined since taking the product of two integers modulo 2 gives an element in $\mathbb{Z}/2\mathbb{Z} \cong \mathbb{Z}_2$.

It is linear since

$$\begin{aligned} B(2m + 2m', n) &= (2m + 2m')n \mod 2 \\ &= 2mn + 2m'n \mod 2 \\ &= B(2m, n) + B(2m', n) \end{aligned}$$

in $\mathbb{Z}/2\mathbb{Z}$. The argument for linearity in the second coordinate and for scalar multiplication from $\mathbb{Z}$ on the coordinates is analogous.

Then

$$\beta(2 \otimes 1) = 1 \neq 0.$$

Thus there is a nonzero element sent to 0, so

$$\ker(\iota_R) \neq 0.$$



Problem 2.4. Let M be an R -module and let $A, B \leq M$ be submodules. You are given

$$A \cap B, A, B, A + B.$$

Construct a natural short exact sequence involving all.

Sol. We note that

$$A + B \supseteq A \supseteq A \cap B$$

and similarly,

$$A + B \supseteq B \supseteq A \cap B.$$

Thus there are very natural inclusion maps

$$A \cap B \rightarrow A, A \cap B \rightarrow B, A \rightarrow A + B, B \rightarrow A + B.$$

However, these are not what we are looking for exactly.

The observation that

$$\frac{A + B}{B} \cong \frac{A}{A \cap B}$$

gives us what we need.

Let $\iota : A \cap B \rightarrow A$ be the inclusion, which is well-defined since $A \cap B \subseteq A$.

Then define $\pi : A \rightarrow (A + B)/B$ to be the map

$$\pi(a) = a + B.$$

This feels like the projection map since for $a + b$ in $A + B$ we have $(a + b) + B = a + B$ since $b \in B$.

It is not terribly hard to see that these are homomorphisms.

Then

$$\text{im}(0 \rightarrow A \cap B) = 0 = \ker \iota,$$

and

$$\text{im } \iota = A \cap B = \ker \pi$$

since the $A \cap B \subseteq A$ go to 0 modulo B , and

$$\text{im } \pi = \frac{A+B}{B} = \ker(A+B/B \rightarrow 0),$$

since π is surjective: take $(a+b) + B \in (A+B)/B$, then

$$(a+b) + B = a + B$$

since $b \in B$, which is mapped to by $a \in A$.

Thus

$$0 \rightarrow A \cap B \xrightarrow{\iota} A \xrightarrow{\pi} \frac{A+B}{B} \rightarrow 0$$

is exact, reaffirming the fact that, for exact sequence

$$0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$$

we have

$$C \cong \frac{B}{\text{im}(A \rightarrow B)}.$$

Another natural route is to look at $\psi : A \oplus B \rightarrow A+B$ by

$$\psi(a, b) = a + b.$$

Then the kernel forces

$$\begin{aligned} a + b &= 0 \\ a &= -b \end{aligned}$$

so $a, b \in A \cap B$. This points to the natural map $\varphi : A \cap B \rightarrow A \oplus B$ by

$$\varphi(x) = (x, -x).$$

Then the sequence

$$0 \rightarrow A \cap B \xrightarrow{\varphi} A \oplus B \xrightarrow{\psi} A+B \rightarrow 0$$

is exact.

This is not hard to prove as

$$\text{im}(0 \rightarrow A \cap B) = 0 = \ker \varphi$$

since $(x, -x) = 0$ iff $x = 0$. Then

$$\text{im } \varphi = \{(x, -x) \mid x \in A \cap B\} = \ker \psi$$

by the earlier argument. Lastly

$$\text{im } \psi = A + B = \ker(A + B \rightarrow 0)$$

since ψ is surjective (for $a + b \in A + B$ there is $(a, b) \in A \oplus B$). ■

Problem 2.5. Let M be an R -module and let $r \in R$. Consider only

$$M, \text{Ann}_M(r) = \{m \in M \mid rm = 0\}, M/rM.$$

There is a very natural endomorphism

$$M \rightarrow M$$

associated with r .

Discover an exact sequence involving

$$\text{Ann}_M(r), M, M, M/rM.$$

Solution and proof. A natural endomorphism on M associated with r could be $\rho : M \rightarrow M$ by

$$\rho(m) = rm.$$

Since modules are closed under multiplication from the ring, this is well defined. It is not terribly hard to see that it is indeed a homomorphism since

$$\begin{aligned} \rho(am_1 + m_2) &= r(am_1 + m_2) \\ &= ram_1 + rm_2 \\ &= a(rm_1) + rm_2 \\ &= a\rho(m_1) + \rho(m_2). \end{aligned}$$

This map need not be injective, and it need not be surjective either. Take for instance R to be $\mathbb{Z}$ and M to be $\mathbb{Z}/2\mathbb{Z}$ with $r = 2$. Then the image does not include $\bar{1}$.

The first useful fact is that for $m \in \text{Ann}_M(r) \subseteq M$ we have $m \xrightarrow{\rho} 0$.

Which gives a candidate:

$$0 \rightarrow \text{Ann}_M(r) \xrightarrow{\iota} M \xrightarrow{\rho} M \xrightarrow{\pi} M/rM \rightarrow 0.$$

Let $\iota : \text{Ann}_M(r) \rightarrow M$ be the inclusion map, which is well-defined since $\text{Ann}_M(r) \subseteq M$.

Then let $\pi : M \rightarrow M/rM$ be the projection map

$$m \mapsto \overline{m}.$$

Then

$$\text{im}(0 \rightarrow \text{Ann}_M(r)) = 0_{\text{Ann}_M(r)},$$

but $\text{Ann}_M(r) \subseteq M$ so

$$0_{\text{Ann}_M(r)} = 0_M,$$

hence

$$\text{im}(0 \rightarrow \text{Ann}_M(r)) = \ker \iota.$$

Next, we have that

$$\text{im } \iota = \text{Ann}_M(r).$$

By definition, for $m \in M$,

$$mr = \rho(m) = 0$$

if and only if

$$m \in \text{Ann}_M(r),$$

so

$$\ker \rho = \text{Ann}_M(r) = \text{im } \iota.$$

Next

$$\text{im } \rho$$

is $rM = \{rm \mid m \in M\}$, but this is just kernel $\ker \pi$ since $x = 0$ in M/rM if and only if $x \in rM$ if and only if

$$x = rm, \quad m \in M.$$

Thus

$$\text{im } \rho = \ker \pi.$$

Lastly,

$$\text{im } \pi = M/rM$$

as the projection is always surjective since there is at least one representative $m \in M$ for each residue class modulo rM .

Now we take a look at a concrete example with $R = \mathbb{Z}$, $M = \mathbb{Z}/12\mathbb{Z}$ and $r = 8$.

Then

$$\text{Ann}_M(8) = \{\overline{m} \in \mathbb{Z}/12\mathbb{Z} \mid m \cdot 8 = n \cdot 12, \quad n \in \mathbb{Z}\}.$$

. The annihilators (worked out concretely further down) are then

$$\overline{0}, \overline{3}, \overline{6}, \overline{9},$$

meaning

$$\text{Ann}_M(8) = (3).$$

Then

$$8M$$

is the residue classes modulo 12 after being multiplied by $8 = 2 \cdot 2 \cdot 2$. Concretely,

$$\begin{aligned} 8 \cdot \bar{0} &= \bar{0} \\ 8 \cdot \bar{1} &= \bar{8} \\ 8 \cdot \bar{2} &= \bar{4} \\ 8 \cdot \bar{3} &= \bar{0} \\ 8 \cdot \bar{4} &= \bar{8} \\ 8 \cdot \bar{5} &= \bar{4} \\ 8 \cdot \bar{6} &= \bar{0} \\ 8 \cdot \bar{7} &= \bar{8} \\ 8 \cdot \bar{8} &= \bar{4} \\ 8 \cdot \bar{9} &= \bar{0} \\ 8 \cdot \bar{10} &= \bar{8} \\ 8 \cdot \bar{11} &= \bar{4}, \end{aligned}$$

meaning

$$8M = (4)$$

in this scenario.

Note: We could also have worked out the annihilator by observing that

$$8m \equiv 0 \pmod{12}$$

if and only if

$$12 \mid 8m$$

if and only if

$$3 \mid 2m$$

if and only if (because 2 and 3 are coprime)

$$3 \mid m.$$

For the maps we have that the 0 map has image $\bar{0}$, with the kernel of the inclusion from the annihilator being $\bar{0}$ as well.

Furthermore the image of the inclusion, which I will here also denote ι is itself the annihilator

$$(3).$$

The kernel of the endomorphism sending $m \mapsto 8m$ is the elements for which $8m$ is 0,

namely

$$\bar{0}, \bar{3}, \bar{6}, \bar{9}.$$

The image of the endomorphism is

$$8M = (4)$$

and the kernel of the projection π is $8M$. The image of the projection is the whole module $M/8M$ so the last term

$$\xrightarrow{\pi} M/8M \rightarrow 0$$

is exact.

We have that

$$\frac{\mathbb{Z}/12\mathbb{Z}}{(4)} \cong \mathbb{Z}/4\mathbb{Z}.$$

You can show this by taking the surjective homomorphism $\varphi : \mathbb{Z}/12\mathbb{Z} \rightarrow (4)$ by

$$\varphi(\bar{n}_{12}) = \bar{n}_4$$

and applying the first isomorphism theorem.

For well-definedness we observe that if $n_1 \equiv n_2$ modulo 12 then, since $4 \mid 12$, we have that $n_1 \equiv n_2$ modulo 4.

Is is a homomorphism since

$$\begin{aligned} \varphi(r\bar{n}_{12} + \bar{m}_{12}) &= r\bar{n}_4 + \bar{m}_4 \\ &= r\varphi(\bar{n}_{12}) + \varphi(\bar{m}_{12}). \end{aligned}$$

For surjectivity, observe that

$$\bar{0} \leftarrow \bar{0}, \bar{1} \leftarrow \bar{1}, \bar{2} \leftarrow \bar{2}, \bar{3} \leftarrow \bar{3}.$$

Note: We surpress the earlier used subscripts denoting which modulo we're taking, but the point is it's obvious.

Then we get an exact sequence

$$0 \rightarrow (3) \xrightarrow{\iota} \mathbb{Z}/12\mathbb{Z} \xrightarrow{\times 8} \mathbb{Z}/12\mathbb{Z} \xrightarrow{\pi \circ \varphi} \mathbb{Z}/4\mathbb{Z} \rightarrow 0.$$

■

Problem 2.6. Let $f : M \rightarrow N$ be an R -module homomorphism, and let $A \leq M$ and $B \leq N$ be submodules. Consider the candidate map

$$\bar{f} : M/A \rightarrow N/B, \quad \bar{f}(m + A) = f(m) + B.$$

1. Determine a necessary and sufficient condition on A , B , and f for $\bar{f}$ to be well-defined.
2. Assume that condition holds. Prove that $\bar{f}$ is an R -module homomorphism. Determine

$$\ker \bar{f}$$

as a quotient involving $f^{-1}(B)$ and A , and determine

$$\operatorname{im} \bar{f}$$

as a quotient involving $f(M)$ and B .

3. Specialize to the case

$$A = \ker f, \quad B = 0.$$

Recover the first isomorphism theorem from the formulas above.

4. Specialize instead to

$$A = f^{-1}(B).$$

Analyze the induced map

$$M/f^{-1}(B) \rightarrow N/B.$$

Determine whether it is injective and describe its image.

Proof of 1.

